Rate blindness: five physical-limit audits of high-profile archaeometric claims

[AUTHOR NAME AND AFFILIATION — to be completed] · Contact: labix7@gmail.com · Code and data: [REPOSITORY URL]

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Abstract

Five recent and widely publicised archaeometric claims reason from a *total*, an *equilibrium*, a *capacity* or a statistical relationship used outside its range, while the process they invoke is limited by a *rate*:

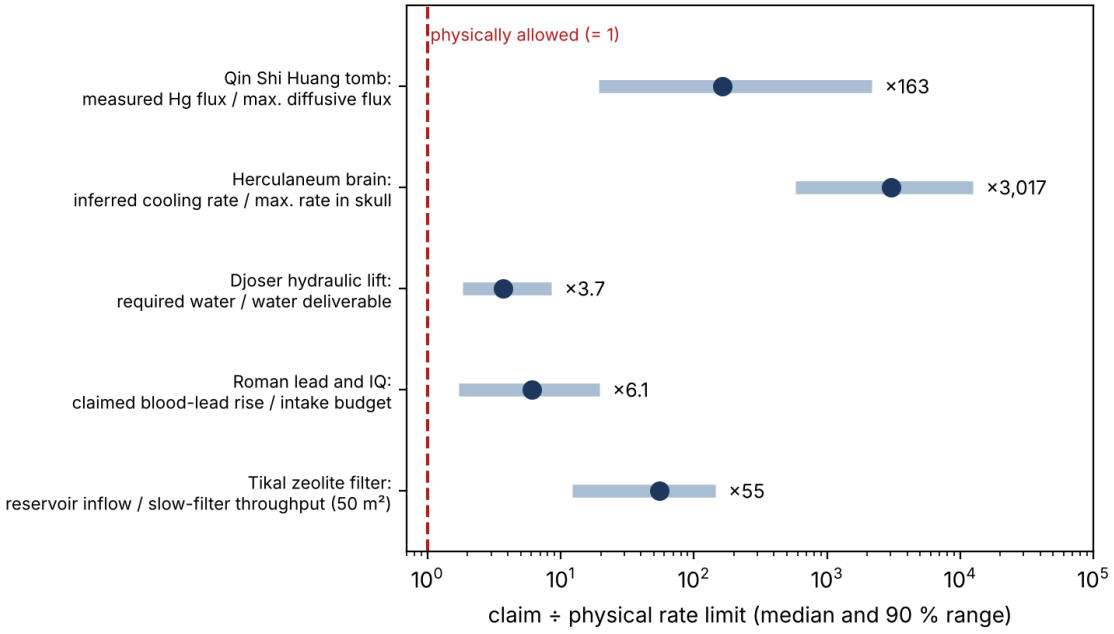
- that mercury vapour from the unopened tomb of Qin Shi Huang escapes through the soil (Duan et al. 2020);
- that a Herculaneum victim's brain vitrified on cooling at ~1000 K s^{-1} (Giordano et al. 2025);
- that the Step Pyramid of Djoser was raised with a flood-fed hydraulic lift (Landreau et al. 2024);
- that airborne lead from Roman mining cost 2.5–3 IQ points across the Empire (McConnell et al. 2025);
- that a zeolite filter purified the drinking water of a Maya reservoir at Tikal (Tankersley et al. 2020).

We call this *rate blindness*. In each case we test the claim against the relevant transport limit, using the authors' own numbers and Monte Carlo propagation of uncertainties. The results:

1. **Qin.** The lidar-measured mercury outflux exceeds the maximum diffusive flux from a saturated chamber by a factor of 19–2,170 (median 163). Only advective exchange can supply it: the tomb would have to "breathe" ~460 m^3 of saturated air per day.
2. **Herculaneum.** The cooling rate inferred from calorimetry exceeds what heat conduction allows inside a skull by a factor of ~600–12,500. At physically possible rates, the authors' own kinetics give a fictive temperature that overlaps the burial temperatures.
3. **Djoser.** Fed by episodic floods into the proposed reservoirs, the lift receives only 27% (12–55%) of the water the authors themselves compute. The binding constraint is storage between floods, not the total volume.
4. **Roman lead.** The air-to-blood regression used gives the *total* blood lead of US children in 1988–94, not an increment, and it is applied 10–100× below its calibration range. Differenced correctly it gives 1.4 IQ points; an intake budget (inhalation, crop deposition, soil) gives 0.6 (0.2–1.8). The paper's 2.5–3 points is 2–5× too high.

5. **Tikal.** A filter slow enough to remove microbes, placed at the reservoir inlet, can pass only ~2% of tropical storm runoff. Treating 90% would need ~6,300 m² of filter bed and ~7,500 t of material carried 30 km.

For each case we give a falsifiable prediction that needs no excavation, or only a sediment core.



Each claim divided by the relevant physical rate limit (median and 90% range of the Monte Carlo). A value of 1 is the largest the physics allows.

1. The pattern

Archaeological science often argues from *stocks*: how much mercury is in a tomb, what glass-transition temperature a material shows, how much water a catchment provides. Stocks are easy to estimate and easy to compare. But each mechanism also needs a *flux* to operate on the stated time scale, and every flux has a ceiling:

- diffusion is limited by $D \cdot \Delta C / L$;
- conductive cooling is limited by $k/(\rho c \cdot L^2)$;
- a reservoir fed by episodic events is limited by storage capacity, overflow and losses.

When the required rate exceeds the ceiling, the conclusion fails however large the stock is. The test is cheap: an order-of-magnitude transport calculation with the paper's own geometry and parameters. In the three cases below it changes the conclusion.

2. Qin Shi Huang: a tomb that cannot be sealed

Claim. Duan, Zhang & Svanberg (Sci. Rep. 10, 10387, 2020) measured atomic mercury above the mound by differential-absorption lidar. They estimated an outflux of $\approx 5 \times 10^{-8}$ kg s^{-1} (uncertain by a factor of ~2) and attributed it to mercury escaping from the burial chamber, "possibly through cracks". The chamber ceiling lies ≈30 m below the surrounding ground, under a ≈50 m mound.

Test. Mercury vapour above liquid mercury saturates at $C_{sat} \approx 9.4$ mg m^{-3} at 15 °C. Steady diffusion through L = 30–80 m of compacted loess, with Millington–Quirk diffusivity (porosity 0.30–0.45, water saturation 0.3–0.8) and a source area of 4,000–15,400 m², gives at most 3×10^{-10} kg s^{-1} (90% range 3×10^{-11} to 2×10^{-9}). This bound ignores adsorption, which only lowers the flux further. The measured flux is **163× larger (19–2,170×)**, and none of 2×10^5 Monte Carlo draws closes the gap. Even a free-air channel spanning the whole roof falls short in 88% of draws. Typical emission from the mound soil (~190 ppb Hg; 10–100 ng m^{-2} h^{-1}) also falls short, by a factor of 9–220.

Implication. If the lidar flux is right and the source is the tomb, the underground palace must exchange air with the atmosphere at ~460 m³ day^{-1} (145–1,460). Barometric pumping of the palace volume can supply this. Using the measured 2016 pressure record (Jinghe station, 27 km away), the supply is marginal in summer, mostly through the daily pressure tide, and larger in winter.

Predictions (pre-registered in a git-timestamped repository):

- emission rises when barometric pressure falls;
- soil-gas mercury increases with depth;
- winter emission exceeds summer emission;
- a pre-registered night-by-night ranking of the 2016 campaign.

3. Herculaneum: a cooling rate heat cannot deliver

Claim. Giordano et al. (Sci. Rep. 15, 5955, 2025) measured DSC onsets of 420 °C at 0.42 K s^{-1} and 510 °C at 1000 K s^{-1} on glassy fragments (0.5–5 mm) “sampled intact, directly from within the skull”. They inferred a natural $T_f \approx 510$ °C and cooling “at rates comparable with our experimental 1000 K s^{-1}” once a short-lived ash cloud dissipated. For the rate they cite cooling of tephra falling through air.

Test.

- *Cooling.* Transient conduction in a heated head (7 cm brain plus 6.5 mm bone, initially at 560 °C, convective and radiative loss to ambient) gives 0.1–1.7 K s^{-1} as each point crosses 510 °C. That is **~600–12,500× below** the inferred rate. Even free fragments in still air reach only 17–720 K s^{-1} (5 to 0.5 mm).
- *Fictive temperature.* At physically possible rates, the paper’s own kinetics (E = 390 kJ mol^{-1}) give $T_f \approx 405$–440 °C. This overlaps the 315–465 °C burial deposits, the very criterion the authors used to reject 420 °C.
- *Survival.* With the same E, the material would soften within 0.4 s at 465 °C and anneal in minutes to hours at 315–350 °C. The present calorimetric signal would record the burial, not the cloud.
- *Heating.* Drying and heating the brain above 510 °C needs ≥3–40 min of full exposure, which is marginal for a cloud lasting “minutes”.

Prediction. Measuring the natural T_f with the enthalpy area-matching method, instead of equating it with a heating-rate onset, will give ≈400–440 °C or less.

4. Djoser: enough water in total, nowhere to keep it

Claim. Landreau et al. (PLOS ONE 19, e0306690, 2024) propose that a float in the 7 × 7 m north shaft raised the stones. Their own figures are:

- water needed: up to 18 Mm^3 over 20–30 years;
- supply from the Wadi Taflah: 4–54 Mm^3 over the same period, which they judge sufficient for intermediate values;
- storage: the Gisr el-Mudir basin (220,000–440,000 m^3, 1–2 m deep) and the Deep Trench (>14,000 m^3).

They state that the figures "do not account for water losses caused by infiltration and other factors". Evaporation is not mentioned.

Test. We ran a daily water balance over the construction period. Floods arrive as a Poisson process at 0.5–5 per year, with the authors' mean supply. The reservoir loses water to overflow, evaporation (1.5–2.7 m yr^{-1}) and infiltration (0.1–3 m yr^{-1}), and feeds the lift through the Deep Trench. The results:

- the lift receives **27% (12–55%) of the authors' own demand**, and none of 1,500 runs reaches 90%;
- even with zero evaporation and infiltration it receives only 37%;
- about half of every flood overflows the basin, and a full basin dries out in 4–30 months;
- the scheme becomes viable (88–99%) only with three things together: a sealed reservoir of Mm^3 scale (for example the whole Dry Moat, 3.5 Mm^3, 20 m deep), optimised lift operation, and ≥1–2 floods per year for decades.

Prediction. If the lift operated, laminated lacustrine silts dating to the 3rd Dynasty will be found at the base of the Dry Moat or the Deep Trench. Flood gravels or aeolian fill alone would refute the only viable water scenario.

5. Roman lead: a regression's intercept counted as pollution

Claim. McConnell et al. (PNAS 122, e2419630121, 2025) combine Arctic ice-core lead with atmospheric modelling and obtain +1.04 ng m^{-3} of air lead over Europe. They convert this to blood lead with BLL = exp[1.932 + 0.140 ln(PbA)], taking the coefficients from Richmond-Bryant et al. (2014; NHANES III, US children 1988–94, an unadjusted ln–ln model). The result, 2.0–5.3 µg dl^{-1} (mean 2.4), is treated as an *enhancement* above a separately estimated Neolithic baseline of 1 µg dl^{-1}. It then becomes a deficit of 2.5–3 IQ points.

Test. - *What the regression predicts.* It gives the total blood lead of children exposed to all sources then present (paint, dust, diet). With almost clean air (0.03 ng m^{-3}) it already predicts 1.6 µg dl^{-1}. - *Extrapolation.* Its calibration medians are ≥0.011 µg m^{-3}, so Roman air is 10–100× below the calibrated range. At that level the implied slope is 355 µg dl^{-1} per µg m^{-3}, against established aggregate slopes of ~2–9. - *Correct difference.* Taking BLL(background + ΔC) − BLL(background) for a natural background of 0.01–0.1 ng m^{-3} gives +1.0 µg dl^{-1} and 1.4 IQ

points. - *Intake budget*. Inhalation, deposition on food crops and soil enrichment, all using the paper's own >500 kt deposited, together with standard child slope factors, give +0.39 µg dl^{-1} (0.12–1.4) and 0.6 IQ points (0.2–1.8).

6. Tikal: a filter that cannot keep up with tropical rain

Claim. Tankersley et al. (Sci. Rep. 10, 18021, 2020) report zeolite in the Corriental reservoir sediments (~58,000 m^3). They propose, in a "hypothetical scheme", an inlet filter of zeolite and quartz sand that "removes harmful microbes" and toxins. The filter sits behind dry-laid walls, is periodically "ejected into the reservoir during flash floods", and is made of material from a tuff ~30 km away.

Test. We generated stochastic hourly rainfall (1,500 mm yr^{-1}) over a 5–50 ha catchment with runoff coefficient 0.3–0.7. A filter of area A_f passes at most $v \cdot A_f$. With slow-sand rates (0.1–0.4 m h^{-1}), the regime that removes pathogens, a 50 m^2 inlet filter treats 1.8% (0.7–8%) of the annual runoff. Treating 90% needs 6,300 m^2 (1,340–16,800) and ~7,500 t (1,650–20,800) of imported material, rebuilt after each flood. A rapid filter (5–10 m h^{-1}; ~220 m^2, ~250 t) is feasible but only clarifies the water.

7. Discussion

All five papers are careful about their measurements; the gap lies in the step from measurement to mechanism. In each case a rate ceiling, fixed by geometry and textbook transport coefficients, was not checked:

- diffusion through soil (Qin);
- conduction out of a skull (Herculaneum);
- storage between episodic floods (Djoser);
- intake by an exposed child (Roman lead);
- the throughput of a filter (Tikal).

The audits change the conclusions in specific ways:

- the Qin tomb is probably not sealed;
- the Herculaneum glass did not record a 1000 K s^{-1} quench;
- the Djoser lift needs a reservoir that has not been identified;
- the Roman IQ effect is 2–5× smaller than claimed;
- the Tikal filter could have clarified the water, but not purified it.

The selection is not random. We screened ten widely reported claims with a quick rate check. Five passed it or could not be decided with public data (the ScanPyramids Big Void, which passed a detector-position test; Teotihuacan muography; Mt Echia muography; the Hierapolis Plutonion; and X-ray dating of the Turin Shroud, which shows an internal tension that depends on unmeasured humidity effects), and five failed. A failure rate of five in ten among prominent papers suggests the omission is common rather than exceptional.

Each audit also yields a cheap, decisive test. We suggest that a one-page “rate check” become routine for mechanism claims in archaeometry. The recipe: identify the flux the mechanism requires, compute its physical ceiling with the paper’s own geometry, and report the ratio.

Limitations.

- Each audit uses simplified geometry and literature ranges for transport properties. The conclusions rest on the size of the mismatches (one to four orders of magnitude for Qin and Herculaneum) rather than on precise values.
- For Djoser the mismatch is a factor of ~2–9 and depends on the palaeoclimate, so we present the result as conditions for viability rather than a refutation.
- For Roman lead the effect is reduced, not eliminated. Exposure near the mines may have been much larger.
- For Tikal the catchment and the hourly rainfall are regional ranges, not data from the paper.
- The Qin result is conditional on the lidar flux.

Data and code

All calculations, Monte Carlo inputs and figures are open:

- Qin: `simulador/qin_mercurio.py` and `simulador/qin_presion_prerregistro.py`;
- Herculaneum: `simulador/herculano_vidrio.py`;
- Djoser: `simulador/saqqara_balance_hidrico.py`;
- Roman lead: `simulador/plomo_romano.py`;
- Tikal: `simulador/tikal_filtro.py`;
- results: `resultados/42–47_*.json`.

The pre-registered predictions are P15–P16 in `PREDICCIONES.md`.

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